DAIO Defense AI Observatory

Thinking Before Sinking

Red and Blue AI Tactics for Naval Combat

Heiko Borchert, Christian Brandlhuber, and Jeronimo Dzaack

DAIO Study 25|27

Ein Projekt im Rahmen von

dtec.bw Zentrum für Digitalisierungs- und Technologieforschung der Bundeswehr

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About the Defense AI Observatory

The Defense AI Observatory (DAIO) at the Helmut Schmidt University in Hamburg monitors and analyzes the use of artificial intelligence by armed forces. DAIO comprises three interrelated work streams:

- Culture, concept development, and organizational transformation in the context of military innovation
- Current and future conflict pictures, conflict dynamics, and operational experience, especially related to the use of emerging technologies
- Defense industrial dynamics with a particular focus on the impact of emerging technologies on the nature and character of techno-industrial ecosystems

DAIO is an integral element of GhostPlay, a capability and technology development project for concept-driven and AI-enhanced defense decision-making in support of fast-paced defense operations. GhostPlay is funded by the Center for Digital and Technology Research of the German Bundeswehr (dtec.bw). dtec.bw is funded by the European Union – NextGenerationEU.

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Content

1 Summary

Mayday, mayday. Can you hear us? We are sinking.
Hello, this is the German Coast Guard.
We are sinking! We are sinking!
What are you "thinking" about?[1]

Are you sinking or thinking? The introductory quotation is from the legendary 2006 Berlitz ad produced by Bates United, Oslo. As you'd expect from a company that offers language courses the ad perfectly illustrates that speaking foreign languages can be a matter of life or death. But jokes aside, there is more to the question than just the thick accent of the German coast guard officer. Rather, the question goes to the heart of how best to apply military power to achieve your goals.

Readers of our previous papers[2] know that we seek to demonstrate that artificial intelligence (AI) learns (= thinking) to develop complex tactics to achieve military missions (= sink, in this case). In *Thinking Before Sinking* we demonstrate that AI develops complex offensive tactics (red) for speed boat swarms without predefined human planning or parametrization. We also illustrate that AI-based policies improve the target sequencing of blue force defense solutions like close-in weapon systems (CIWS), which is illustrated by higher mission success rates in our simulation runs.[3] Finally, we show that AI-based CIWS coordination improves blue mission success. Simulation results indicate that AI-based coordination provides military value beyond massing fire power. Our collaborative coordination tactics excel the more complex the military coordination challenge grows[4] and thereby significantly increase the effectiveness and efficiency of available self-defense systems.

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1 https://www.youtube.com/watch?v=WjUPIZhqhDs (last accessed 11 May 2025).
2 Borchert/Brandlhuber/Brandstetter/Schaal, Free Jazz on the Battlefield; Borchert/Brandlhuber, His Hands Can't Hit What His Eyes Can't See.
3 In this regard our paper also contributes to the ongoing debate about whether new technologies will tilt the balance towards the offensive and defensive.
4 However, our findings also suggest that this won't be enough, for example, for convoy protection missions to defend high-value platforms. These missions will require additional and intelligent stand-off weapons such as sea-launched effectors, for example, to ensure protection.

These findings matter because naval forces not only increasingly face the resurgence of naval peer-to-peer conflicts. Technological advancements also provide sub-peers the opportunity to challenge well-established navies as well as commercial shipping. In today's threat environment, blue's high-value naval platforms easily become red's high value targets. Contested naval theaters and limited loading capacities of surface platforms create challenges for the optimal armament of navy ships. "Commanders likely err towards greater capability relative to expected missions in selecting defensive interceptor loadouts, increasing the cost of interceptors on board," noted RADM Kavon Hakimzaeh, Commander of the US Carrier Strike Group 2, upon reflecting on initial lessons identified from naval combat in the Red Sea.[5]

His comment reminds us of the importance of diligent planning ahead of naval missions to gauge the optimal armament in response to presumed adversarial threats. As our paper shows, sophisticated simulation leveraging AI can contribute to this task by illustrating how different effector combinations can be optimized for successful countermeasures. In addition, his statement emphasizes the economic dimension of warfighting. High value assets are expensive. The annual operating cost of a US carrier like the Dwight D. Eisenhower "is about USD657.4 million" without air wing costs, and medium-range, surface-to-air missiles like the SM-2 Block IIIC, for example, "cost about USD2.5 million apiece."[6] Economy of effort in using these assets matters, and AI that advances economy of effort via collaborative coordination greatly enhances mission effectiveness and efficiency.

This, in turn, is relevant to tackle logistical challenges. In response to changing maritime threats, navies emphasize the need for dispersion and distribution. While distributing capabilities across different assets makes it more difficult for red forces to neutralize blue forces, distribution also poses new questions regarding supply security. This ties into the broader debate on contested logistics, i.e. the ability of red forces to disrupt blue force supply lines.[7] Although our findings are preliminary, they seem to present AI tactics as a possible way out of the conundrum; assets that are more effective in countering threats are likely to alleviate the security of supply burden.

5 Quoted in: Fabey, "Lessons learnt. US Naval operations in the Red Sea," p. 28.
6 Fabey, "Lessons learnt. US Naval operations in the Red Sea," p. 27.
7 For more on this concept, see: Hughes, "Giving our 'paper tiger' real teeth."

In sum, the exploratory red and blue naval AI tactics discussed in this paper provide the ground for GhostPlay@SEA, the next phase of a capability and technology development project started in 2021 (see box below). GhostPlay@SEA blends into the new vision of the German Navy, which emphasizes the need for advanced readiness and presence in relevant theaters, the use of adequate weapon systems to repel adversarial aggression, as well as the use of advanced technologies for uncrewed assets and AI to master high-risk scenarios.[8] In this context, GhostPlay@SEA will transfer collaborative machine-learning and reinforcement learning tactics from land-air constellations to the naval domain. The goal is to expand the use of these tactics from domain-specific swarms of uncrewed systems to their multi-domain use and to prepare for attacks with uncrewed assets across different domains by developing multi-domain counter swarm tactics.

This is the third paper in a series addressing the role of AI tactics on the battlefield. These papers grow out of GhostPlay, a capability and technology development project aimed at developing red and blue AI tactics. *Free Jazz on the Battlefield*, our first paper published in 2022, laid out the core idea, described key aspects of the basic method – for example regarding the use of decentralized and non-hierarchical command and control – and presented our initial findings focusing on the role of air defense against aggressor swarms. *His Hands Can't Hit What His Eyes Can't See*, the second white paper published in September 2024, discussed the use of red and blue AI tactics for swarms of air-launched effects against ground-based air defense systems. This paper presented collaborative coordination as a key mission requirement and discussed its use for fully autonomous reconnaissance-strike swarm tactics.

8 Das Zielbild für die Marine ab 2035, pp. 2-3, 6.

2 Rough Seas Ahead

Reflections on the Naval Conflict Picture and Its Consequences

The maritime domain is fragile. Fragility results from competing interests of great powers, emerging powers and sub-peers as well as violent non-state actors that use the sea as a means of transportation, resources, habitat, and a vector to project power.[9] As the former Development, Concepts, and Doctrine Center (DCDC) of the UK Ministry of Defense argues, the future operating environment, that also shapes naval missions, is going to be more contested, cluttered, congested, connected and constrained.[10] Recent conflicts in the Red Sea and the Black Sea reemphasize some of these elements and thus provide a (partial) glimpse into the likely future naval conflict picture. The following preliminary findings are of particular relevance for our analysis:

- **Dynamized Land-Sea Interactions**
  The use of uncrewed surface vehicles (USV) and uncrewed aerial vehicles (UAV) in combination with missiles has been dynamizing land-sea interactions. In the littorals, which constitute the focus of our analysis, distance hardly provides a sanctuary. Ukraine, for example, has significantly "extended" its shoreline by hitting Russian targets at 300 nautical miles with land-based forces using USV and UAV.[11] Similarly, Iran and Houthis have shown they are able to hit ships at significant range using UAV and missiles in combination.[12] While it is true that well-trained and well-equipped Navies can successfully counter these threats – as the US Navy has shown in the first-ever use of anti-ship ballistic missiles by the Houthis[13] – this may not be the case for lightly defended warships or merchant ships.[14]

- **Resurgence of Naval Air and Missile Warfare**
  Current conflicts forcefully underline the need to significantly invest in naval air and missile warfare capabilities. Uncrewed systems, anti-ship cruise missiles (ASCM) and anti-ship ballistic missiles (ASBM) constitute formidable threats for surface platforms that are very likely to drive growth for protective systems like close-in weapon systems (CIWS).[15] Current systems face the challenge that sensing solutions and effectors need to be readjusted to concurrently defend larger areas, that require longer range effectors, and specific point targets, that require shorter-range interceptors.[16]

9 Borchert, "Maritime Sicherheit in Gefahr," pp. 54-65.
10 Future Character of Conflict, pp. 21-25.
11 Parker, "The Black Sea battle." To what extent unmanned systems used in Ukraine would be fit for mission against, for example, the People's Liberation Army Navy (PLAN) is questionable. See: Tallis, "The calm before the swarm."
12 Sutton, "9 Lessons from Iranian and Houthi Attacks on Ships in the Red Sea."
13 Ziezulewicz, "What Red Sea battles have taught the Navy about a future China fight."
14 Sutton, "9 Lessons from Iranian and Houthi attacks on ships in the red sea;" Kushal, "Lessons for the Royal Navy's future operations from the Black and Red Sea."
15 Karako/Pyle, "Operations in the Red Sea: Lessons for Surface Warfare," p. 7; Dunley, "The end of the age of transoceanic navies?," p. 61.
16 Fabey, "Lessons learnt: US Naval operations in the Red Sea," p. 28.

- ### Contested Logistics

  The need for increasingly varied effector portfolios when operating close to contested enemy shorelines requires concept developers and planners to put more emphasis on novel solutions to deal with contested logistics. For one, this is likely to prompt the need for different logistics hubs at sea such as auxiliary vessels, multi-role support ships – although these would offer attractive targets – and new combinations of manned and uncrewed support platforms.[17] In addition, the value of mass may be limited as it is "constrained by how much platforms can carry."[18]

- ### Cross-Domain Interaction

  At sea the crew cannot focus on one threat only. Rather the above findings reiterate the cross-domain character of operations close to shore. Naval combat in the Black and Red Seas also underlines the value of third-party information. In the first half of 2024, for example, the United States provided Ukraine with satellite-based intelligence, surveillance, and reconnaissance data more then 30 times a day. In a similar way, Russia has been sharing satellite data with the Houthis to attack merchant ships.[19] Data sharing, however, requires adequate data-sharing frameworks. UK Royal Navy destroyers, for example, "do not currently possess a cooperative engagement capability or the ability to download data via (multifunction advanced data links) the way the US vessels equipped with Aegis baseline 9 can."[20] In addition, incoming engagement data needs to be assessed. In the Red Sea, the US Navy showed that what used to take two weeks has been done in two days, but this required "a lot of bandwidth to go back and forth."[21]

- ### Adaptability

  Finally, the above findings point towards adaptability as a key requirement. To operate successfully in hybrid combat against uncrewed systems in the Red Sea, for example, the US Navy had to "tweak existing systems and tactics, techniques, and procedures."[22] Among other things, this led the US Navy to use existing assets in different ways: "Using Hellfire against USV, air-to-air, aviation platform shooting down UAVs."[23] These and other tweaks underline the value of battlefield engineering, with experts being able "to take data from weapon systems to learn and improve tactics" to respond to novel aggressor behavior.[24]

17 Kaushal, "Lessons from the Black and Red Sea on the use and design of future fleets,". Expeditionary reload at sea is a most challenging undertaking, but can be done. See Fabey, "Lessons learnt: US Naval operations in the Red Sea," p. 25-27.
18 Tallis, "The calm before the swarm."
19 Kaushal, "Lessons for the Royal Navy's Future Operations from the Black and Red Sea;" Fabey, "Lessons learnt: US Naval operations in the Red Sea," p. 25, 29.
20 Kaushal, "Lessons from the Black and Red Sea on the use and design of future fleets."
21 Fabey, "Lessons learnt: US Naval operations in the Red Sea," pp. 25, 29.
22 Ibid., p. 24.
23 Pomerleau, "CNO Franchetti: 'We're continuing to learn' countering drones, missiles in the Red Sea."
24 Pomerleau, "CNO Franchetti: 'We're continuing to learn' countering drones, missiles in the Red Sea;" Karako/Pyle, "Operations in the Red Sea," p. 9.

How, you may ask, should concept developers respond to these and other challenges? Among the different concepts currently under development, in which distributed forces play a prominent role, Mosaic Warfare is most relevant for our approach. Mosaic Warfare reinterprets force design and the Observe-Orient-Decide-Act (OODA) loop originally conceived by John Boyd.[25] Prominent among military planners, most armed forces focus on accelerating and compressing the loop to leverage decision superiority vis-à-vis adversaries. In so doing, most initiatives focus on optimizing joint situational awareness and understanding (observe, orient) as well as precision engagement (act), whereas the core decision-making process receives less attention.

This is the angle that Mosaic Warfare, developed by the US Defense Advanced Research Projects Agency (DARPA), is focusing on. The concept assumes that successful military operations require comprehensive decomposition of existing force packages into small units that will be reconfigured at campaign time and commensurate with mission requirements and challenges presented by enemy forces (Table 1).[26] Consequently, command and control (C2) relationships "follow communications availability, rather than attempting to build a communications architecture that supports a desired C2 structure."[27] Thus, Mosaic Warfare proposes a non-hierarchical and largely decentralized approach to decision-making[28] meant to "impose multiple dilemmas on an enemy."[29]

Implementing Mosaic Warfare puts a premium on adaptability, which features prominently in the UK Royal Navy's idea of creating a "Protean Maritime Force." At the core of the Protean force is the ability to improvise and innovate to deal with uncertainty.[30] Consequently, a Protean power emphasizes "emergence, not linearity or regularity" as a key principle of successful adaptation.[31] As Mosaic Warfare suggests, such a Protean force will "shift in form and focus depending on the demands of the circumstances." Rather than optimizing "highly specific platforms" this concept emphasizes a "force with interdependent and interchangeable systems components" – and thus goes beyond equipment-centric ideas to "encompass the philosophical and intellectual foundations of the force."[32]

25 For more on John Boyd, his concept, and its reception, see: Hammond, The Mind of War; Robinson; The Blind Strategist.
26 Gryson, "Mosaic Warfare," p. 11.
27 Clark/Patt/Schramm, Mosaic Warfare, p. V.
28 Borchert/Brandlhuber/Brandstetter/Schaal, Free Jazz on the Battlefield, pp. 16-17.
29 Gryson, "Mosaic Warfare," p. 11.
30 In Greek mythology, Proteus, "the first-born of Poseidon (…) changed shape as soon as he was seen." Katzenstein/Seybert, "Preface," p. xi.
31 Borchert, "The Very Long Game of Defense AI Adoption," p. 7.
32 Maritime Operating Concept, pp. 19, 10.

Table 1: From Kill Chains to Mosaic Warfare

	Distributed Kill Chain	System-of-Systems	Adaptive Kill Web	Mosaic Warfare
Approach	Manual integration of existing systems	Systems prepped for multiple battle configurations	Semi-automated ability to select a pre-defined effects web prior to mission	Ability to compose new effects webs at campaign time
Benefits	■ Extends effective range ■ Increase engagement opportunity	■ Enables faster integration and more diverse kill chains ■ Greater effect possible by removal of co-location equipment	■ Allows pre-mission adaptation ■ More lethal, imposes complexity on adversary	■ Adaptable to dynamic threats and environment ■ Scaling to many simultaneous engagements
Challenges	■ Static ■ Long to build ■ Difficult to operate and scale	■ Limited ability to scale ■ Cannot add new capabilities on the fly	■ Static "playbook" ■ Limited number of kill chains ■ May not scale well	■ Scaling limited by human decision makers

Sources: Gryson, "Mosaic Warfare,", p. 11; Clark/Patt/Schramm, Mosaic Warfare, p. 29.

In addition to adaptability, a look at adversarial concepts reveals another important design principle: temporal elasticity. Western ideas tend to emphasize speed, as in the hyperwar[33] concept, to overwhelm adversaries. Adversaries, by contrast, seek to outmaneuver their challengers "either by moving slowly or deniably to avoid provoking a response (…) or by moving quickly in areas where policies are unclear and potential responses are too slow or ineffectual (…)."[34]

33 Allen/Husain, "On Hyperwar."
34 Dougherty, "Moving Beyond A2/AD."

Combining a nuanced approach on the temporal dimension with the ideas of Mosaic Warfare and the Protean force provides the basis for a new paradigm. This strives to exploit temporal latitude by maximizing the time available to make decisions rather than simply compressing the OODA loop. As discussed elsewhere,[35] the mathematical framework of Markov Decision Processes (MDP) and their extensions to partially observable or continuous time cases provide one of the most suitable approaches to implement this ambition. Ultimately this approach will decompose a sequential OODA loop by delegating O-O-D-A tasks to those nodes in a federated effects web best suited to execute the respective task at any given point in time. The decomposition will significantly advance parallel processes because each sub-cycle will be baked into the effects web thus enabling the execution of multiple and independent but coordinated O-O-D-A sub-cycles across the effects web. Consequently, the OODA cycle will "hide" in the network making it almost impossible for adversaries to identify and detect and thus deny and degrade decision-making. As we will discuss later, this new paradigm underpins the logic of collaborative (self-)coordination.[36]

35 Borchert/Brandlhuber/Brandstetter/Schaal, Free Jazz on the Battlefield, p. 13.
36 Borchert/Brandlhuber, His Hands Can't Hit What Hit Eyes Can't See, pp. 28-30.

3 Black Pearl

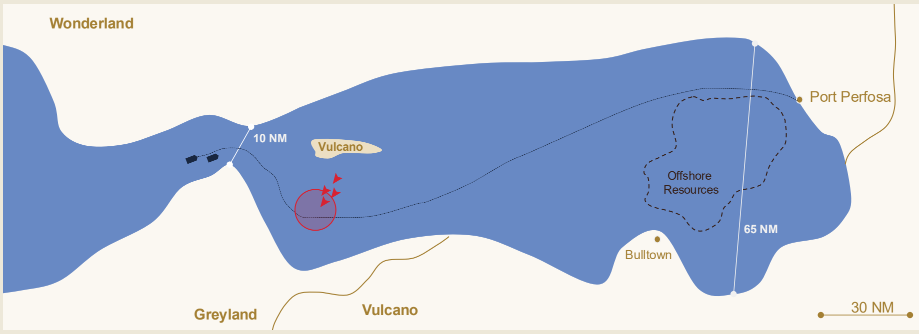
Adjusting the Defense Metaverse for Naval Combat

The Black Pearl is the fictious ship under the command of Jack Sparrow in Pirates of the Caribbean. It provides the codename for a feasibility study that we have conducted – and summarize in this paper – to explore if red and blue AI tactics, that have been developed for land-air missions, can be transferred to the naval domain.[37] The goal was to develop stochastically optimal tactics to defend a high value target – like the Black Pearl – against a swarm of aggressor speed boats.

3.1 Scenario

Wonderland expects its ally Blueland to provide strategically relevant goods at sea. Its tanker is entering the Long Sea from the West as illustrated in Figure 1. Blueland also provides a frigate for convoy protection and to ensure Wonderland's unrestricted access to open sea lanes of communication. Vulcano, Wonderland's neighboring archrival, has been interrupting Wonderland's in- and outbound sea traffic for many years. Volcano Island, strategically located close to the opening of the Long Sea, provides an ideal basis to conduct naval harassment missions. In so doing, Vulcano is using manned and uncrewed swarms of speed boats to interrupt and delay Wonderland's commercial shipping routes and inflict economic damage by prompting higher freight costs. On a regular basis, Vulcano also executes aggressive missions to board tankers and attack naval platforms sent into the theater to protect commercial vessels.

Figure 1: Tanker Protection Mission



37 Such a domain transfer is challenging as the lack of cover and concealment and a higher impact of material factors tend to make naval battles faster and more decisive. For more, see: Biddle/Severini, "Military effectiveness and naval warfare."

3.2 Red and Blue Models

As discussed above, swarms of speed boats constitute a significant threat for commercial vessels and naval platforms. For Black Pearl we have developed red coordination tactics by modelling speed boat swarms according to the details summarized in Table 2. Specific environmental aspects have not yet been modelled, for example, by dedicated wave models. However, specific effects that result from sea state changes or wind have been implicitly considered via the probability hit rate.

Table 2: Red Force Platform Characteristics

Platform	Length (meters)	Beam (meters)	Draught (meters)	Propulsion	Speed (knots)	Sensors	Weapon systems
Tondar Fast Attack Craft	38.6	6.8	2.7	3 diesel engines 3 shafts	35	SR-47A surface search radar Type 341 fire control radar	2 twin missile launchers (Qader AShM) 2 twin AK-230 guns 2 twin 23mm guns
Seraj Speed boat	15.5	4.4	.75	2 diesel engines	75	None (except navigation)	12.7mm machine gun 107mm 12-barrelled MRL
Peykaap Missile Boat	17.3	3.75	0.7	2 diesel engines Propeller	52	EO Fire control radar	2 single AShM 2 324mm torpedo tubes
Tareg (Bohammer RL-118/130) Speed boat	13	2.7	0.7	2 diesel engines 2 shafts	46	Surface search radar	RPG7 rocket launcher 12.7mm machine gun 106mm recoilless rifle 107mm 12-barelled MRL
Shahid Nazeri (HARTH 55)	55	14.1		2 shafts	38	Unknown	Remotely controlled 20mm three-barrel rotating cannon

Abbreviations: AShM Anti-Ship Missile, EO: Electro-Optical sensor, MRL Multiple Rocket Launcher

Sources: "Tondar-class fast attack craft," "Seraj-class speed boat," "Peykaap III-class missile boat," "Taregh-class speed boat," "IRIS Shahid Nazeri."

Given the exploratory nature of our study, our blue force modelling efforts focus on coordinating effectors onboard its naval ships. To this purpose we modelled 27mm close-in weapon systems (CIWS)[38] and 76mm guns. We have used ballistic models with linear and quadratic air-drag to model the naval gun and implemented a simplified probabilistic damage model (Bayesian model) combined with a terminal ballistic model. The base version of Blueland's frigate has two CIWS, one on each side. Onboard installations and the positioning of both systems limit the frigate's lateral alignment range to +/- 100 degrees. In addition, the frigate operates a 76mm gun at the front of the platform.

Today, CIWS are remotely operated typically using electro-optical sensors and laser rangefinders. Electro-optical sensors use zooms up to x48. In addition, optical tracking mechanisms ensure concurrent tracking of multiple targets to reduce sensor-effector latency when switching targets. To engage fast and agile swarms, optimal target sequencing is needed to prevent aggressor swarms from exploiting their attack positions.

In most cases, the electronic combat management systems (CMS) or human operators provide for the target sequencing. We have used a model-based reinforcement learning approach for AI-based target sequencing and coordinate two or more CIWS with decentralized partially observable Markov Decision Processes (decPOMDP) that take into account decisions taken by the partner CIWS in the decision-making process.[39]

38 In the scenarios used for this paper, the CIWS' caliber is 27mm x 145 with a cadence of 1,700 rounds per minute (v0 1,100m/s). The system's elevation is -15 degrees to +60 degrees with azimuth +/- 170 degrees and a maximum angular acceleration of around 85 degrees per second.

39 Brandlhuber, "Superiority on the Battlefield with Tactical AI and Emergence."

3.4 Training AI Tactics

Black Pearl uses the Defense Metaverse, which has been partially developed within GhostPlay, as the core simulation environment.[40] In addition to providing a highly realistic replica of the battlefield, the Defense Metaverse uses counterplay to let the AI learn how to develop tactics that "are best suited to achieve mission success within given specific parameters set by humans."[41]

Given the exploratory nature of Black Pearl, we have used a particle simulation for initial training runs. Contrary to intuition, transferring reinforcement learning strategies (policies) from the simulation environment to the real world does not work particularly well if the initial training takes place in high-resolution simulators. Quite often, the respective strategies show instabilities as so-called spurious effects – subtle effects that may be artifacts of the simulation environment, but which the system judges to be relevant for decision-making – tend to affect the training. Therefore, a more promising approach is to learn the basic decision-making process on relatively coarse resolution simulators that are subject to noise (e.g., particle simulations) and to use high-resolution simulators to finetune the respective policies.

In addition, we use a multi-agent approach to simulate naval warfare scenarios including intelligent adversaries. This implies that we do not only model single entities but also groups of entities that need to be coordinated. Realistic simulation requires joint situational awareness and understanding as well as decentralized coordination. Decentralized coordination implies that individual entities can implement individual strategies and cooperate with other entities if cooperation is more beneficial to achieve a joint mission.

As cooperation is not the default mode of AI-based agents, specific methods and mechanisms are needed to provide incentives for AI-based agents to learn how to cooperate. Multi-agent reinforcement learning is even more demanding and requires specific modifications of the training process. With Independent Q-Learning, for example, each AI-based agent "independently learns its own policy, treating other agents as part of the environment."[42] Using Centralized Training with Decentralized Execution (CTDE) agents train using generally available information, but implement self-learned strategies only based on local observations.[43]

40 Borchert/Brandlhuber, His Hands Can't Hit What His Eyes Can't See, pp. 12-14.
41 Ibid, p. 12.
42 Foerster, "Stabilizing experience replay for deep multi-agent reinforcement learning," p. 1.
43 Amato, "An introduction to centralized learning for decentralized execution in cooperative multi-agent reinforcement learning."

The Defense Metaverse used for the Black Pearl simulation models coordination for multi-agent systems as a collaborative game using a model-based reinforcement learning approach solving a decPOMDP.[44] The resulting models offer several advantages:

- First, agents using decPOMDP are required to develop strategies that do not only focus on their own state and actions. Rather, they need to consider probable conditions, intentions and actions of fellow agents, which – in turn – provides the ground to self-learn collaborative strategies.
- Second, decPOMDP handle uncertainty and partially observable facts. Thus, agents learn how to develop strategies with incomplete information. These strategies are most useful where ground truth is missing, i.e., in situations where information is sparce and latency delays decision-making. Consequently, these strategies can compensate for gaps in a Recognized Maritime Picture (RMP).
- Third, agents trained with decPOMDP are required to evaluate the benefits of individual action versus collective action. This capability is paramount when joint mission success depends on individual action such as swarming.
- Finally, operating in a decPOMDP regime requires AI-based agents to learn effective communication strategies. Agents need to learn, when and what to communicate thereby considering available bandwidth and the impact of information sharing on collective decision-making. This, for example, is most important when operating under adversarial electromagnetic fire.

44 Borchert/Brandhuber/Brandstetter/Schaal, Free Jazz on the Battlefield, pp. 13, 17.

4 Emergent AI Tactics for Naval Combat

4.1 Red Force Tactics

We use AI-based policies to implement the scenario elements such as the naval assets as well as the respective sensors and effectors. These policies are based on distributed coordination. As each element has its own control module there is no central red force control. Rather, as discussed in the previous chapter, each element follows its own strategy. Using this foundational principle, each agent is using training rounds to develop specific policies, which provide for the optimal use of the physical characteristics and the specific sensor and effector capabilities simulated in the training runs. As a result, different attack tactics emerge in response to blue force actions and countermeasures.

Against this background, Figure 2 illustrates the starting point of the scenario. As discussed in chapter 3.1, the Panamax tanker (dotted blue line) enters the theater with a frigate providing convoy protection. A group of aggressor boats, illustrated by the red dots, is loitering in the Northeast. An initial batch of aggressor boats – consisting of two Peykaap speed boats and the bigger High-Aspect Rajo Twin Hull (HARTH) boat – approach the convoy as illustrated in Figure 3.

The red arrow shows the interception mission of the speed boats that approach the tanker at 40 to 50 knots. The speed boats block the convoy. In response, the frigate is accelerating to get a better understanding of enemy actions in adjacent areas and to neutralize the threat – if needed – before the red group reaches the tanker. At the same time a second group of aggressor boats carrying heavy weapons is heading towards the convoy (Figure 4).

Figure 2: **Scenario Starting Point**

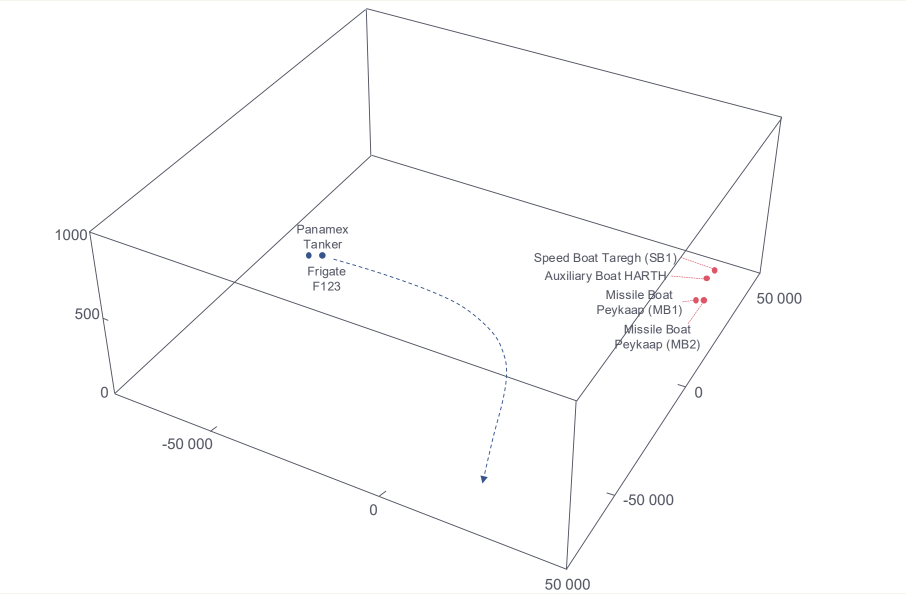


Figure 3: **First Aggressor Boats Approach the Convoy**

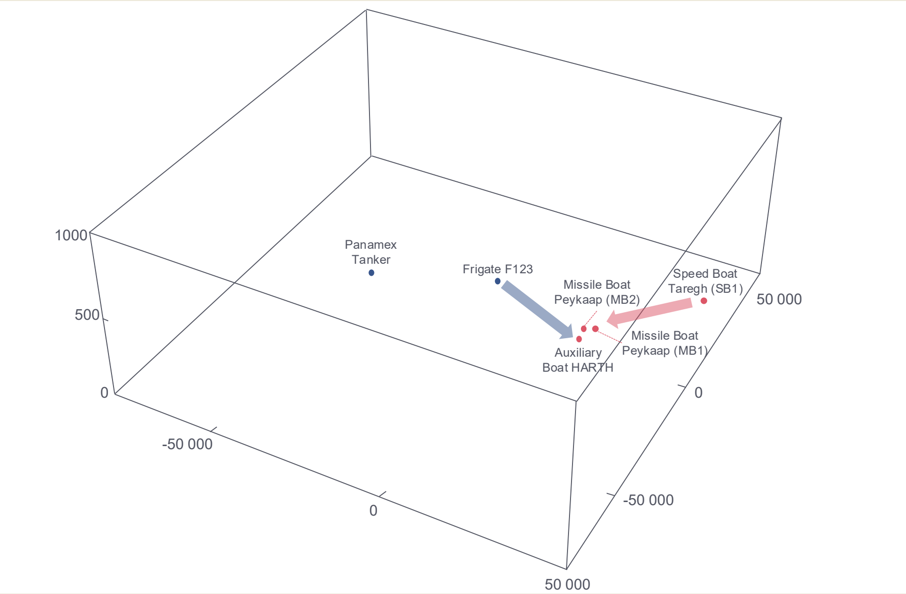
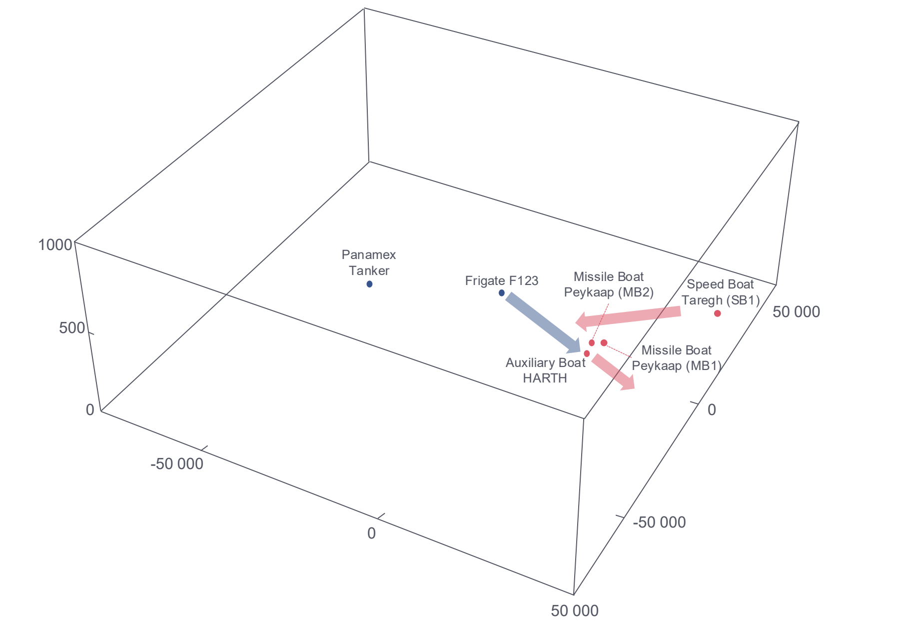


Figure 4: Aggressor Boats Trying to Intercept the Tanker



The second aggressor group is adjusting and synchronizing attack vectors in relation to the first batch as well as the convoy. As can be seen in Figure 4, the speed boats try to separate the tanker from the frigate with additional speed boats attempting to lurk the frigate away to create a gap that other red swarm members will seize to attack the tanker. As the initial response of the frigate to counter the second batch of aggressor boats sets in, a third group of red speed boats is heading towards the convoy. One of these speed boats could be used to board the tanker while another high-speed member of the third batch carries artillery rockets to engage the convoy. Figure 5 shows the blue frigate's RMP and illustrates attack trajectories and timing of the aggressor swarm. The blue dot in the middle symbolizes the position of the frigate, the other blue dot to the left indicates the position of the tanker.

As the scenario continues the boats currently blocking the tanker are moving to the North to evade engagement with the frigate. In parallel, the fast missile boats reposition themselves also in the North to threaten the frigate with missiles and torpedoes. This is meant to deter and delay the frigate's engagement of the second and third aggressor batches (Figure 6).

Figure 5: Blue Frigate's Recognized Maritime Picture

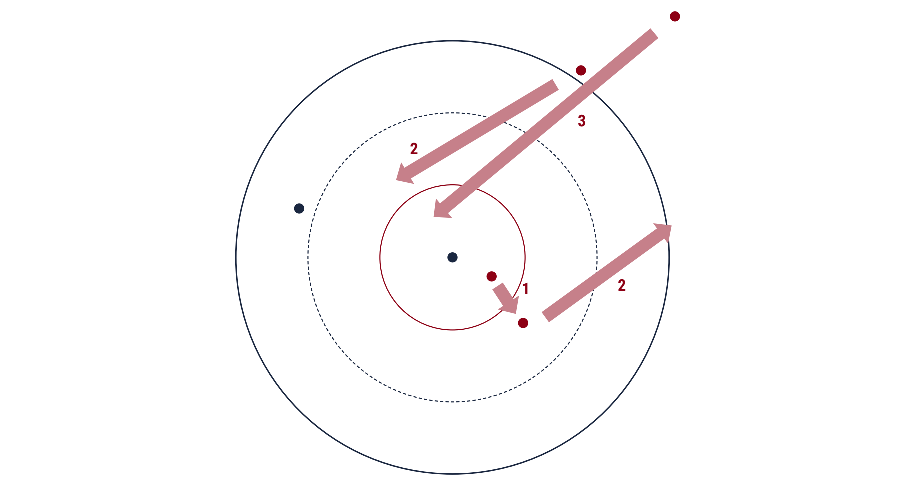
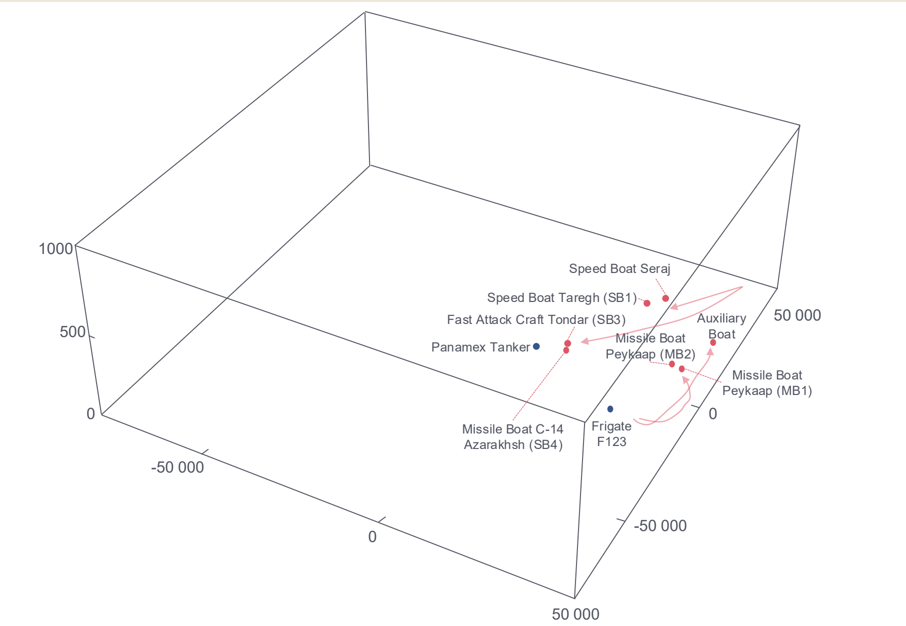
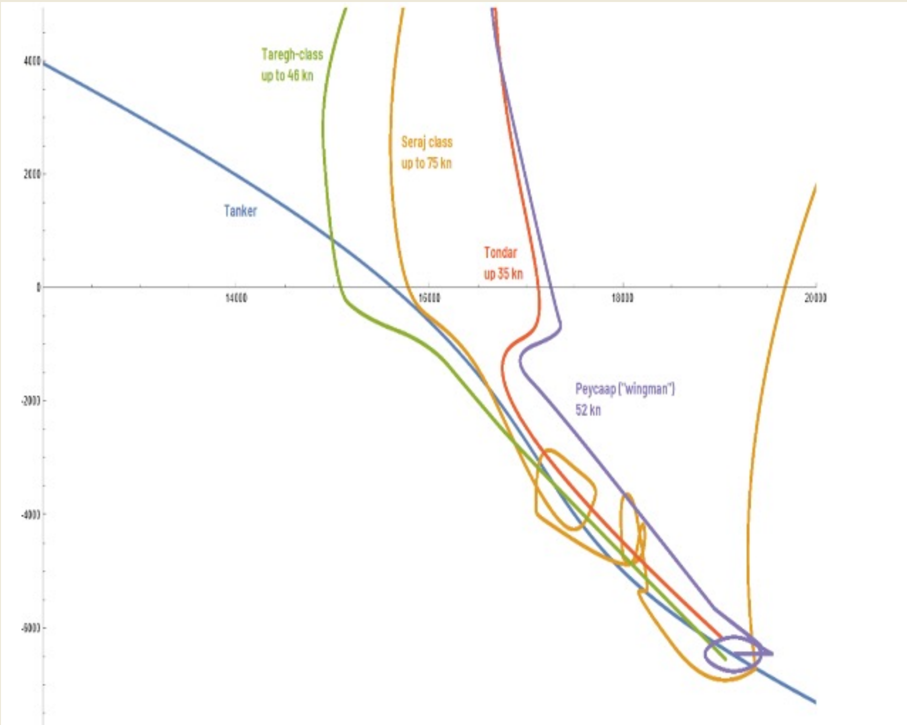


Figure 6: Aggressor Swarm Blocks Tanker and Repositions Itself



What is this scenario telling us? The trajectories (Figure 7) show that aggressor speed boats learn to develop a tactic that enables them to coordinate and synchronize complex fast boat attacks with armed and unarmed platforms. The Taregh class trajectory (green) and the Seraj class trajectory (orange) illustrate that these swarm members leverage speed to position themselves in front of the tanker. They also circumvent the tanker and cross the tanker's trajectory to force the tanker to reduce tempo and ultimately come to a halt. In parallel the Tondar class boats (red) and the Peykaap class boats (purple), which are equipped with heavy weapons, move side by side with the tanker or position themselves behind the tanker to limit the tanker's freedom of maneuver. In some of the scenarios, the aggressor boats were allowed to use weapons; in these cases, the onboard gun systems of the Tondar and Seraj class boats were activated, not to destroy the tanker but to enable the crew to board the tanker. At the same time, the Peykaap class boat trajectories (purple) also illustrate that these boats would have been able to destroy the tanker with torpedoes at any time.

Figure 7: Red Intercept Tactics and Trajectories



Against this background, our findings, first, show that the above swarm coordination tactics emerge without central coordination. Based on the principles described elsewhere,[45] each swarm member perceives its peers as part of the environment and can asynchronously exchange coordinating information with them.

Second, each agent has developed a policy that makes optimal use of the platform's characteristics. Speed boats, for example, individually and collectively leverage tempo and acceleration for fast repositioning to encircle the tanker, cross the tanker's trajectory, evade frigate engagements and optimally position their weapons in relation to the tanker and the frigate. In addition, Seraj class boats use speed to increase the reach of onboard artillery rockets and to create a buffer zone vis-à-vis the frigate.

Third, red force timing is gradually maturing. We observe that red force swarms learn to adapt their timing in response to the behavior of the blue frigate.[46] Time sequences between individual actions of the swarm were expanded, which enabled swarm members to reposition at high speed. Our assessment illustrates that this tactic is meant to enable the swarm to better evade the frigate's coordinated weapons engagement – which, in turn, reduces the effectiveness of blue force countermeasures.

4.2 Blue Force Tactics

Given the scope of our exploratory analysis, blue force tactics focused on ensuring collaborative coordination between the frigate's navigation trajectory and the coordination of two CIWS and the 76mm naval gun onboard the frigate. Both CIWS operate starboard and backboard around 6 meters above sea-level, with each system covering only one area of the ship. The high above-sea level weapon position created room for attackers to break or sneak through the weapon's defense perimeter and reach the hull of the frigate. AI-powered platforms thus need to coordinate the frigate's path of navigation and weapon system activities in a way that addresses threats created by speed boats while keeping attackers within the effective weapon range.

As we were mainly interested in behavioral aspects of red and blue platforms, we assumed perfect perception (ground truth) for red and blue forces in the simulation. Projectile trajectories have been explicitly computed using a second order

45 Borchert/Brandlhuber/Brandstetter/Schaal, Free Jazz on the Battlefield, pp. 16-17.
46 This behavior is similar to what we have observed with aerial aggressor swarms attacking ground-baed air defense systems. See: Borchert/Brandlhuber, His Hands Can't Hit What His Eyes Can't See, p. 27.

ballistic model as we wanted coordination policies to consider aspects like mechanical latency of turning and stabilizing the weapon as well as cadence, reloading and cool-down of cannons. We estimated actual hit rates under environmental conditions (e.g., wind) and potential damage to the targets using a probabilistic graphical model.[47]

Against this background we note, for example, that improved tactical target sequencing can substantially improve blue mission success. Today, existing CIWS track, classify and identify multiple targets within their sensors' observation angle. Sequencing is typically provided based on CMS heuristics. This, however, introduces latency and often makes insufficient use of actual information about each weapon state. Sequencing optimized with AI, however, can further improve CIWS effectiveness and efficiency, as our findings for heterogeneous aggressor swarms show. In this case target sequencing considers the trajectories of the swarms and the specific threat each object poses for the tanker-frigate convoy. During training, policies also develop a notion of how the situation may unfold in the future given past encounters with speed boats in previous training runs. The blue policy seems to be able to preempt and disrupt assaults thus successfully crippling attacks very early on. Therefore, as Figure 8 shows, the CIWS identified Seraj class speed boats carrying artillery rockets as the number one threat (blue arrows) that needs to be countered before addressing Taregh class boats, whose weapons are of limited reach (orange arrows). After engaging these platforms, the third engagement round focuses on the remaining Taregh class boats (green arrows).

Sequencing also needs to be seen in tandem with the coordinated use of the weapons against the aggressor swarm. In one specific scenario, we witnessed the very early engagement of the swarm by the frigate. Contrary to the scenario discussed in the previous chapter, the frigate did not address the speed boats that were harassing the tanker, but instantly targeted coordinated fire at a Peykaap boat and its support vessel (Figure 9). Consequently, the second Peykaap boat lost cover and quickly left the theater to regroup. Although the early use of the blue CIWS is escalatory, this move prevents the tanker from being separated from the frigate.

As the threat of the first speed boat group has been neutralized, the frigate can turn around and head towards the remaining boats that attack the tanker. This puts the aggressor swarm under pressure. As each speed boat's sensors are mounted within line of sight, a direct attack of the frigate is getting more difficult. When within reach, the frigate engages the speed boats that maneuver in front of

47 It is important to note that at this stage of the project we did not use classified performance and measurement data. Thus, we parametrized environmental and damage models using hit and damage probabilities that seem plausible but are not based on live sampled data. This suggests that real world hit and damage probabilities can deviate from simulated findings. However, the model we have used can be easily replaced to rerun training runs and reevaluate the findings based on real data.

Figure 8: CIWS Sequencing Tactic Against Heterogeneous Aggressor Swarms

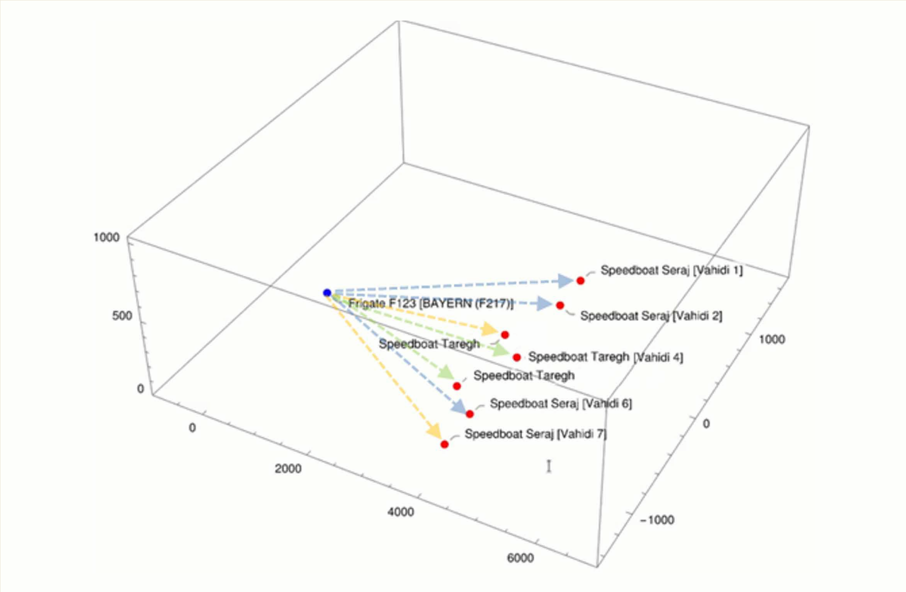
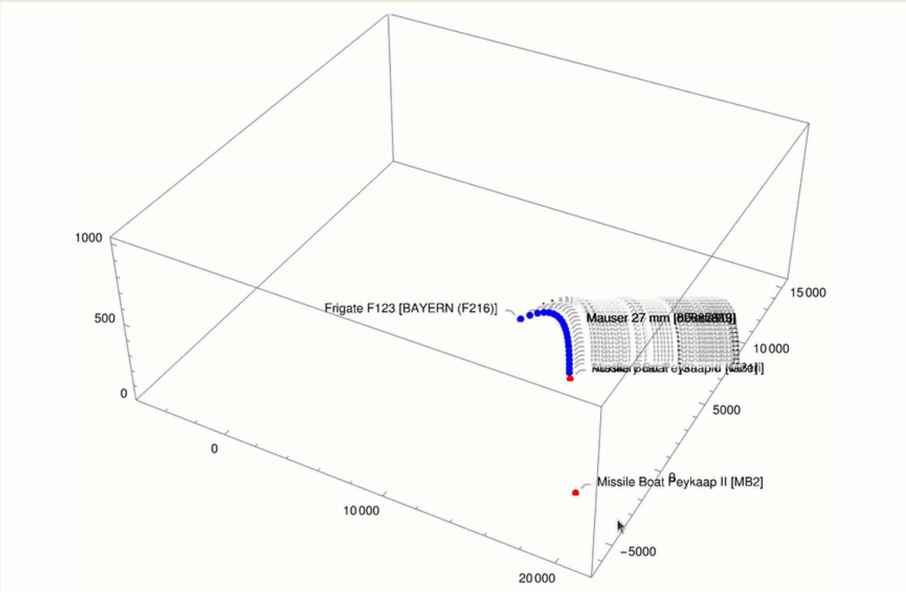


Figure 9: Attacking the Peykaap Boats Early On



the tanker. In addition, the frigate moves around the tanker and first engages the Tondar class boats carrying torpedoes and anti-ship missiles. Then the frigate engages the Taregh class boarding boats. At this stage the Seraj class boat no longer constitutes a threat for the tanker as it is too close to the tanker to effectively use its 107mm artillery rocket and is insufficiently manned to board the tanker (Figure 10). However, a Peykaap class boat, that had been loitering outside the weapon range of the frigate, is now approaching the convoy to support the Seraj class boat. In response the frigate moves into a position that allows for the effective concurrent and coordinated use of both CIWS to successfully attack the Seraj and Peykaap class boats at the same time.

We acknowledge that early escalation is one option for blue to achieve mission success. This is noteworthy as our assessments show that more reactive and defensive postures significantly reduce blue mission success given that red aggressor swarms leverage speed, agility, and range to their advantage. Overall, the red aggressor swarm will effectively neutralize the frigate's protection when the swarm succeeds in (a) separating the tanker from the frigate and (b) confronting the frigate with a situation in which the tanker and/or the frigate are within reach of all speed boat weapons. In this case the Seraj class boats will typically address the convoy at high-speed using their 107mm rockets, which will distribute in a fan-like formation to force the frigate's sensors and effectors into maximal counter-engagement (Figure 11). This is likely to overwhelm the frigate and produces a favorable situation for the Tondar and Peykaap class boats to attack with torpedoes and anti-ship missiles.

Figure 10: **Sequential Engagement of the Aggressor Swarm by the Frigate**

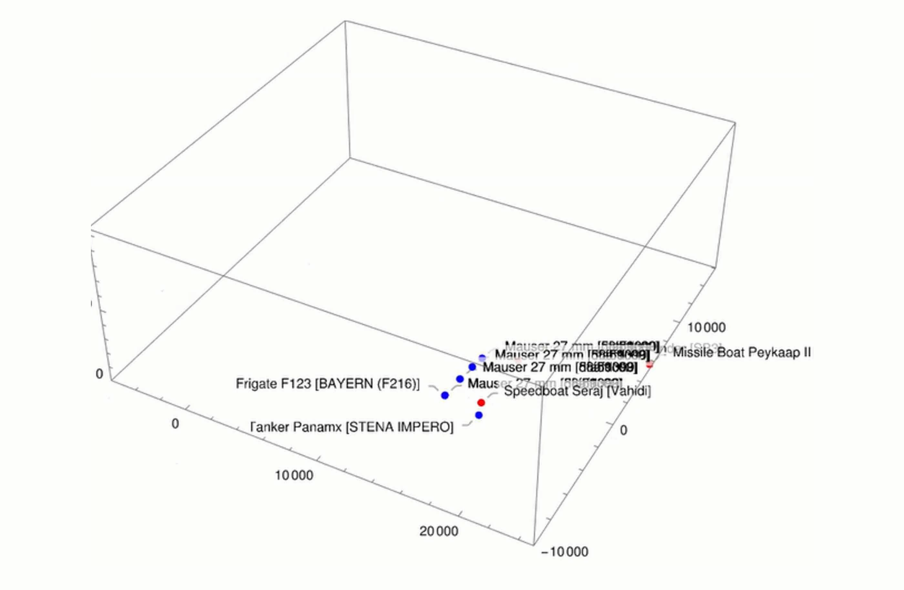
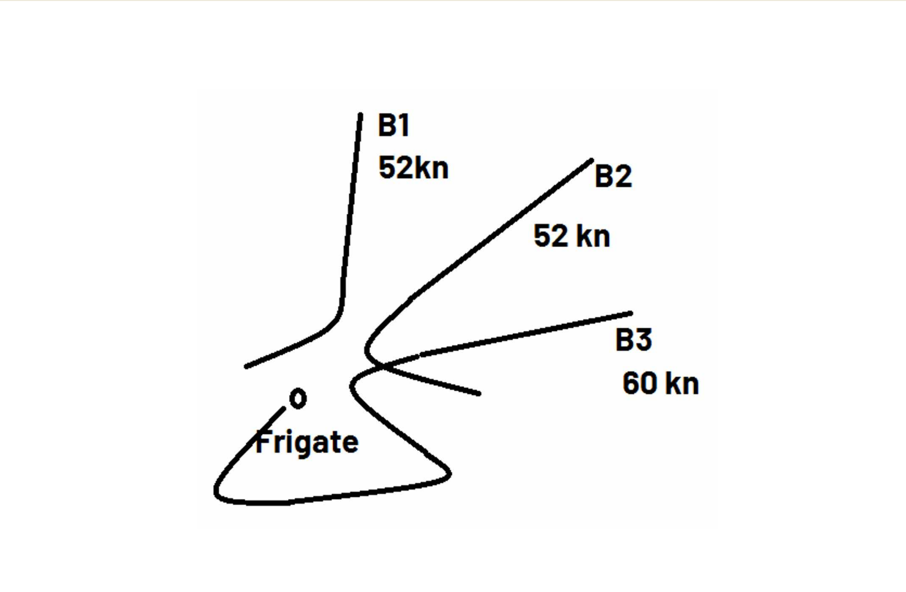


Figure 11: **Typical Attack Pattern of Searj Class Boats Using 107 mm Artillery Rockets**



4.3 Interpreting the Findings

Let us summarize our key findings for red and blue AI tactics based on 100k scenarios. We illustrate two different options. Red Force (free) means that the blue force frigate has been destroyed, that red force elements have boarded the blue force tanker,[48] or that the tanker has been destroyed if boarding was prevented. Overall boarding was weighed with a multiple factor of three. Red Force (restricted) means that red forces cannot destroy the tanker. In particular, red force elements cannot use standoff weapons against the tanker, which suggests that blue force has ample time to regroup and push back the red force swarms. We also compare the coordination tactic for two and four CIWS to gauge the tactic's impact on performance and effectiveness.

Table 3: Statistical Success Rates based on 100k Scenarios

	Number of Successful Scenarios			
	Red Force (free)		**Red Force (restricted)**	
CIWS Coordination	2 CIWS	4 CIWS	2 CIWS	4 CIWS
Traditional	34%	38%	41%	42%
Emergent	39%	52%	56%	72%

Table 3 summarizes our findings, which can be interpreted as follows:

- **No Use of the 76mm Naval Gun**
  The system is quick to learn, that using the 76mm naval gun is not effective in countering the aggressor swarm. In early training cycles the system used the gun to combat groups of aggressor swarms that were operating in close formation or within a corridor. As the systems policies matured, the 76mm naval gun was no longer used.

48 The boarding operation lasted for more than ten minutes with the boarding boat operating next to the tanker.

- **Emergent Coordination**
  Emergent coordination (Figure 12) provides added value even when using two CIWS, but the extra performance is not decisive compared to traditional CIWS coordination mechanisms. This mainly stems from the fact that each system can only cover its backboard or starboard area of responsibility thus preventing both systems from taking red aggressor swarms into the crossfire. Using emergent coordination with four CIWS improves the defender's mission as four systems provide more options for cooperation, which is reflected in the control algorithm.

- **Low Success Rates for Red Force (free)**
  Speed and agility enable red force to cover wide areas without offering blue force effective countermeasures as the impact of the defender system is limited. In the worst case of our modeled scenarios, a single surface platform will protect a high value asset in only 3.9 (two CIWS) or 5.2 (four CIWS) out of 10 cases. Thus, the success rate is rather low, because the defender needs time to reposition vis-à-vis the aggressor while preventing a gap to emerge between the tanker and the aggressor swarm. Consequently, alternative solutions such as sea-launched effects, for example, could be used to cover the gap.[49]

Figure 12: Full-Scale Weapons Engagement of the Frigate Against the Aggressor Swarm

Source: Project BlackPearl by 21Strategies and ATLAS ELEKTRONIK

49 Today helicopters are used to execute the respective mission. But to counter a bigger aggressor swarm that uses Man Portable Air Defense Systems (MANPADS) against the frigate-tanker combination, intelligent sea-launched effects might be more effective.

5 Outlook

Driving Naval AI Tactics Development with GhostPlay@SEA

GhostPlay and GhostPlay@SEA are capability and technology development projects that develop AI tactics to evaluate existing and enable new tactical behavior. In so doing, these projects go beyond the status quo. In contrast to classical AI approaches used for classification and regression, for example to improve image recognition, signal analysis or text transformation, tactical decision-making not only optimizes a single decision at a specific moment in time. Rather, tactical decision-making considers sequential dependencies and adversarial response. What seems necessary and logical at a specific point in time can jeopardize mission success in the long run (myopic decision). In addition, decisions not taken in time can limit the future decision space.

Against this background we have demonstrated that AI can self-learn non-trivial and effective red aggressor tactics for speed boat swarms and blue defender tactics to improve the use of close-in weapon systems (CIWS). In both cases we demonstrate that our decentralized decision-making approach leads to decision policies that enable collaboration. Agents using these policies learn if and when coordination with peer agents may enable them to solve tasks, which cannot be solved by individual agents alone. This leads to effective new assault tactics. Comparative analyses of the blue force indicate that collaborative tactics excel in performance and robustness, especially when the military coordination challenge grows more complex. A yet unsolved challenge, on which we are working, stems from the need to scale the demonstrated approach to substantially larger numbers of coordinated objects.

Although our findings are preliminary given the exploratory nature of the scenario at hand, the message is clear: AI-enabled collaborative coordination tactics constitute a new software-enabled value layer that armed forces can use to advance military effectiveness. Armed forces using a service-oriented architecture to advance defense digitalization will understand these tactics as microservices that can be used to replace monolithic legacy systems – and thus advance modularity, elasticity, and scalability.

This logic underpins GhostPlay@SEA. Based on the AI tactics and the Defense Metaverse developed between 2021-2024, we will spend the next two years to transfer the current technology and tactics stack to the naval domain. In line with conceptual ideas such as Mosaic Warfare and the Protean force, we will use naval AI tactics to advance force distribution. Distribution, in turn, requires coordination by human operators and/or advanced machine-machine cooperation.

It is this latter perspective that GhostPlay@SEA will address to advance lethality with cross-domain swarm tactics and improve defense with cross-domain counter swarm tactics. To this purpose we will combine operational experimentation (OPEX) and simulative experimentation (SIMEX) in a novel way. First, we use the

Defense Metaverse to develop a portfolio of AI tactics, which will be demonstrated in a SIMEX. Second, we strive to transfer these tactics from the digital environment to physical uncrewed assets with the goal to live test the digitally developed tactics during the GhostPlay@SEA OPEX.[50] In parallel, we continue to use the Defense Metaverse to run increasingly complex and demanding scenarios to finetune our AI tactics for the second SIMEX. We intend to demonstrate with this approach that software-defined defense – using concepts and capability development in combination with assessments of the tactical added value of various technological building blocks and digital and real-life experimentation – produces tangible outcomes for force development.

50 Currently, reliable domain transfers constitute a major challenge and are subject to cutting-edge research. The Defense Advanced Research Projects Agency (DARPA), for example, launched a project on "Transfer from Imprecise and Abstract Models to Autonomous Technologies" (TIAMAT) in September 2023. For more, see: https://www.darpa.mil/research/programs/transfer-from-imprecise (last accessed 11 May 2025).

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